

Math 221D Quiz 2 (September 11, 2026)

- (a) (15 points) Find all vectors $\mathbf{x} \in \mathbb{R}^5$ which are orthogonal to $(1, 0, 2, 4, 3)$, $(2, -1, 1, 7, 5)$, and $(-1, 3, 7, 5, 6)$. Express your answer in the form $\text{Span}(\mathbf{v}_1, \mathbf{v}_2)$ for some vectors $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^5$.

Solution. We have

$$\text{ref} \begin{bmatrix} 1 & 0 & 2 & 4 & 3 \\ 2 & -1 & 1 & 7 & 5 \\ -1 & 3 & 7 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}.$$

The solution set of $A\mathbf{x} = \mathbf{0}$ where A is the matrix above is

$$\text{Span}((-2, -3, 1, 0, 0), (1, 0, 0, -1, 1)).$$

- (b) (5 points) Conceivably, your classmates could give a different pair of $\mathbf{v}_1$ and $\mathbf{v}_2$ for part (a) and still be correct. However, show that it is not possible to express your answer to part (a) in the form $\text{Span}(\mathbf{v})$ for a *single* vector $\mathbf{v} \in \mathbb{R}^5$.

Solution. If there's a vector v such that $\text{Span}(v_1, v_2) = \text{Span}(v)$, then in particular, we must have $v_1, v_2 \in \text{Span}(v)$. This means that $v_1 = c_1 v$ for some scalar c_1 , and $v_2 = c_2 v$ for some scalar c_2 . In fact, both c_1 and c_2 must be nonzero, since neither v_1 nor v_2 is the zero vector, so

$$v_1 = c_1 v = c_1 (c_2^{-1} v_2) = (c_1 c_2^{-1}) v_2,$$

i.e., v_1 must be a scalar multiple of v_2 . However, $v_1 = (-2, -3, 1, 0, 0)$ and $v_2 = (1, 0, 0, -1, 1)$, and it is easily seen that they are not scalar multiples of each other.